

Review

The Role of Learning Analytics in Evaluating Course Effectiveness

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Abstract: This study aims to examine the **use of learning analytics in course evaluation within higher education institutions**, in order to identify effective methodologies and best practices for leveraging data to improve educational effectiveness. Following the PRISMA guidelines, a systematic literature search was conducted in Scopus, yielding 34 relevant studies published between 2015 and 2024 for analysis. The results reveal six key categories of learning analytics applications: **sentiment analysis, questionnaire analysis, engagement analysis, topic classification, predictive modelling, and performance analysis**. The data sources for learning analytics applications primarily include questionnaires and learning management systems. While descriptive analysis was found to be the most commonly employed analytical technique, advanced techniques such as machine learning, artificial intelligence, and social network analysis are becoming more prominent. The studies addressed a wide range of elements associated with course evaluation, including course design, content quality, assignments, instructional strategies, workload, feedback mechanisms, and the integration of technology. These findings highlight the importance of adopting holistic approaches to capture the multifaceted nature of student experiences. This study also uncovers major limitations in the existing research, such as small sample sizes, potential biases due to the use of survey-based methods, and challenges in generalising findings across disciplines. These insights underscore the need for further research to enhance the methodologies used in course evaluations. This study contributes to advancing learning analytics practices and emphasises the importance of innovative approaches for evaluating and improving course effectiveness.

Keywords: learning analytics; course evaluation; machine learning; educational data mining; educational quality



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1. Introduction

In an era marked by rapid technological advancements and an increasing emphasis on data-driven decision making, education has been undergoing a transformative shift. Learning analytics, which involves gathering and analysing data about learners and learning contexts [1], has emerged as **a powerful tool to address challenges related to this shift**. By harnessing large amounts of educational data, learning analytics holds significant promise for advancing educational sustainability. It has demonstrated potential in facilitating personalised learning, improving academic performance, and creating adaptive learning environments [2–4], ensuring that educational delivery remains responsive and sustainable while meeting the dynamic demands of society.

As an essential component of course delivery, course evaluation systematically assesses the success and effectiveness of educational offerings [5,6]. It provides evidence regarding

the performance of teaching staff [7], identifies areas for quality enhancement [8], and informs curriculum design and development [9]. Insights gained from course evaluations empower policymakers, course coordinators, and instructors to determine appropriate actions, allocate resources effectively, and implement necessary support measures for continuous course improvement [10].

Course evaluation has traditionally relied on methods such as questionnaire surveys and interviews with students to investigate their perceptions of a course [11]. The integration of learning analytics has broadened its scope, enabling a more data-driven and comprehensive approach. For example, analyses on student engagement data can be conducted regarding participation in discussions, time spent on course materials, video-watching behaviours, and completion of assignments [12–14], providing data-informed insights into how students interact with course contents and activities. Additionally, textual feedback can be processed using natural language processing techniques to uncover common themes, sentiments, and areas of concern [15–17].

Despite the increased adoption of learning analytics in course evaluation, there remains a lack of systematic analyses of the patterns and practices associated with its implementation. This study seeks to address this research gap by examining how learning analytics has been applied in course evaluation within higher education institutions, as well as identifying effective methodologies and best practices for leveraging data to enhance educational outcomes. Specifically, the study is guided by the following research questions:

1. What are the application categories of learning analytics in course evaluation?
2. What are the features of learning analytics in course evaluation, in terms of data collection methods, analysis techniques, and the evaluated aspects?
3. What are the main limitations in existing studies focused on learning analytics in course evaluation?

2. Related Work

Course evaluation involves a systematic assessment of a course's effectiveness in achieving its educational objectives. This process typically encompasses a comprehensive analysis of curriculum design, learning outcomes, and instructional methods [18]. Existing research on course evaluation has addressed a broad range of aspects, such as the purposes of course evaluation [5,19], the elements to assess [20–22], and the stakeholder groups involved in the process [23–25]. Kogan and Shea [26] have proposed a framework for course evaluation that integrates four key dimensions: why evaluate (goals), what is evaluated (process and outcomes), who evaluates (students, peers, and self), and when to evaluate (during the course, end of the course, and sometime in the future).

In higher education settings, student evaluations are the most commonly employed method for course evaluation [11,27]. They typically involve collecting data from course evaluation questionnaires to investigate students' perspectives on the course contents, teaching approaches, instructional methods, and the overall quality of a course. Relevant questionnaires often include both quantitative items in a Likert scale on teaching practices and course components, as well as qualitative, open-ended questions that allow students to provide detailed feedback and suggestions for improvement [28].

Course evaluation is related to but differs from student evaluations of teaching or student satisfaction surveys. Both of the latter aim to gather feedback from students to assess instructional quality and overall learning experiences; however, student evaluations of teaching focus on evaluating the performance and effectiveness of individual instructors, offering targeted feedback on aspects such as clarity of explanation, enthusiasm, availability, and responsiveness to students' needs [29–31]. In contrast, student satisfaction surveys adopt a broader perspective, assessing students' perceptions and feelings about their

educational experience. These surveys cover not only teaching quality and course delivery, but also factors such as administrative support, campus facilities, and extracurricular activities [32]. While high levels of student satisfaction may indicate a positive learning environment, they do not necessarily reflect the educational effectiveness of a course [33]. Furthermore, high ratings for teaching do not always correlate with high ratings for the course itself [34].

Advances in information technology have influenced course evaluation practices, particularly regarding the adoption of online evaluation methods and the evaluation of online courses. Anderson et al. [35] have compared online and traditional paper-based evaluations and identified the advantages of online approaches, including the efficient collection of feedback, the ability to maintain anonymity, and the quick generation and accessibility of evaluation reports. Alturkistani et al. [6] conducted a systematic review on evaluation methods for massive open online courses (MOOCs), highlighting key challenges such as a lack of longitudinal data and the limited consideration of confounding variables.

The vast amount of data generated from online learning and online evaluations has fostered the application of learning analytics for course evaluation. Learning analytics has been applied for tasks such as extracting sentiments and suggestions from students' qualitative feedback [36], analysing students' online learning behaviours and interests [37], and identifying areas for course improvement based on insights from students' performance and learning achievements [38].

Despite the growing use of learning analytics in course evaluation, there has been a lack of comprehensive review studies focused on this emerging area. While individual studies have explored various applications of learning analytics in the context of course evaluation, there is no consolidated understanding of how these tools are being utilised across different contexts and their overall impact. A comprehensive review in this area is needed to provide a holistic view of the current state of research, identify best practices, and highlight areas needing further investigation. To address this gap, the present study examined the empirical literature focused on the application of learning analytics in course evaluation, in order to categorise the applications of learning analytics in this domain and identify the patterns of evaluation practices, including data collection methods, analysis techniques, and the evaluated aspects. This study aims to bridge the existing research gap by offering a detailed synthesis of how learning analytics can enhance course evaluation processes, ultimately contributing to more effective and data-driven educational practices.

3. Methodology

This study investigated how learning analytics contributes to the course evaluation process. Its methodology followed the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) guidelines [39], ensuring a systematic, transparent, and replicable approach to literature review and data synthesis.

3.1. Search Strategy and Screening Method

Relevant research articles were sourced from Scopus. This publication database was selected due to its widely recognised utility for conducting literature reviews across diverse disciplines [40]. In educational disciplines, Scopus has been adopted for reviews in topics related to learning analytics [2–4] and various learning approaches [41,42].

The database search focused on journal articles published between 2015 and 2024 (until 21 October 2024). The following search strings were utilised to query the article titles, abstracts, and keywords:

(“learning analytics” OR “educational data mining” OR “machine learning” OR “artificial intelligence” OR “deep learning”) AND course AND (evaluat* OR measure* OR assess* OR improve* OR enhance*) AND (effectiveness OR quality OR success).

To ensure the selection of relevant studies, the following inclusion criteria were applied:

- i. The study presents empirical research on the application of learning analytics for course evaluation;
- ii. The paper is written in English;
- iii. The full text is accessible;
- iv. Data collection methods and learning analytics techniques are clearly described.

Figure 1 illustrates the search and selection process, which followed the PRISMA framework. Initially, a total of 3726 articles were identified during the database search phase. A rigorous screening process was conducted to eliminate irrelevant articles, during which the titles and abstracts of each paper were scrutinised to verify their alignment with the inclusion criteria. As a result, 3583 records were excluded at this stage. In the subsequent eligibility assessment, 129 records were retained for full-text examination. This step led to the exclusion of 95 records that failed to meet the defined criteria. Notably, articles primarily focused on student satisfaction or student evaluation of teaching were excluded, as these concepts differ substantially from course evaluation, as discussed in the previous section. This selection process ensured a focus on course evaluation and avoided conceptual overlaps. Finally, 34 articles were selected for inclusion in the review.

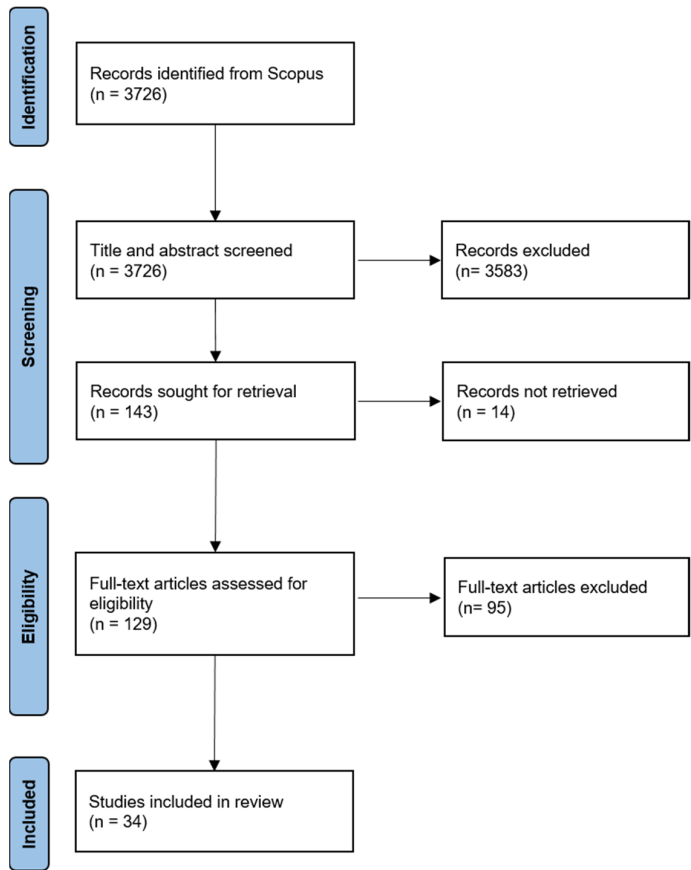


Figure 1. Procedures for the search and selection of relevant publications.

Figure 2 presents the number of studies published annually from 2015 to 2024, highlighting an overall upward trend over the years. The early years saw minimal relevant

activities, with only a total of 4 articles published between 2015 and 2018. However, relevant studies began gaining traction from 2019, culminating in a peak of interest during 2021 and 2022. Although a slight decline was observed in 2023, the sustained interest and ongoing research suggest a promising outlook for 2024. This pattern reflects a growing recognition and exploration of learning analytics as an integral tool for enhancing course evaluation in recent years.

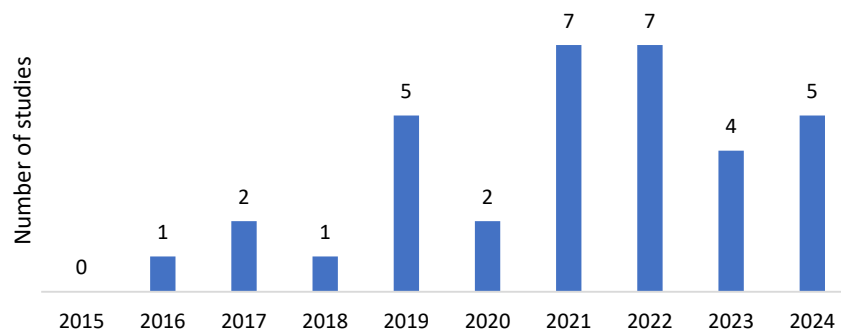


Figure 2. Number of relevant studies in each year from 2015 to 2024.

3.2. Data Analysis

To analyse the 34 selected articles, a content analysis approach was employed. Each article was examined to extract and categorise information on the features of learning analytics approaches used in course evaluation. Based on the analytics techniques reported in the reviewed studies, six categories of learning analytics applications for course evaluation were identified. The analysis focused on (i) the types, sources, and size of data used for learning analytics; (ii) the aspects of courses being evaluated; (iii) the analysis techniques applied; and (iv) the limitations addressed in the studies. The coding process was conducted collaboratively by two researchers to ensure reliability. One researcher was responsible for identifying and categorising relevant information, while the other reviewed and verified the categorisation. Any discrepancies were resolved through further discussion and review until a consensus was reached. Finally, the features of learning analytics approaches for course evaluation were analysed based on the coding results.

4. Applications of Learning Analytics for Course Evaluation

Figure 3 illustrates the six categories of learning analytics applications in course evaluation. Note that some studies utilised multiple applications. The most prevalent category was sentiment analysis, appearing in 14 studies. This was followed by questionnaire analysis with 11 studies and engagement analysis with 10 studies. The remaining categories included topic classification and predictive modelling, each appearing in 7 studies, and performance analysis, which was used in 4 studies.

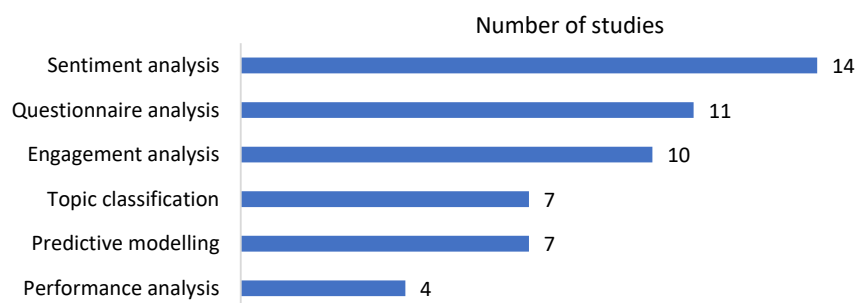


Figure 3. Distribution of studies in each application category.

Figure 4 depicts the adoption of various learning analytics applications across two distinct periods: 2015–2019 and 2020–2024. Despite the number of studies being much larger in the latter period, the proportion of learning analytics applications reveals the focus of development in each period. During the earlier period (2015–2019), engagement analysis was most frequently used, followed by questionnaire analysis. In contrast, the period from 2020 to 2024 marks a significant shift in the types of learning analytics applications used for course evaluation. Sentiment analysis emerged as the most dominant application, underscoring a growing emphasis on understanding student sentiment and its role in improving educational experiences. While questionnaire analysis remained the second most frequent type of application, its growth was more modest compared to the surge in sentiment analysis. The other types of application (engagement analysis, topic classification, predictive modelling, and performance analysis) also showed a notable increase in number, demonstrating comprehensive growth in the use of all types of learning analytics applications for course evaluation.

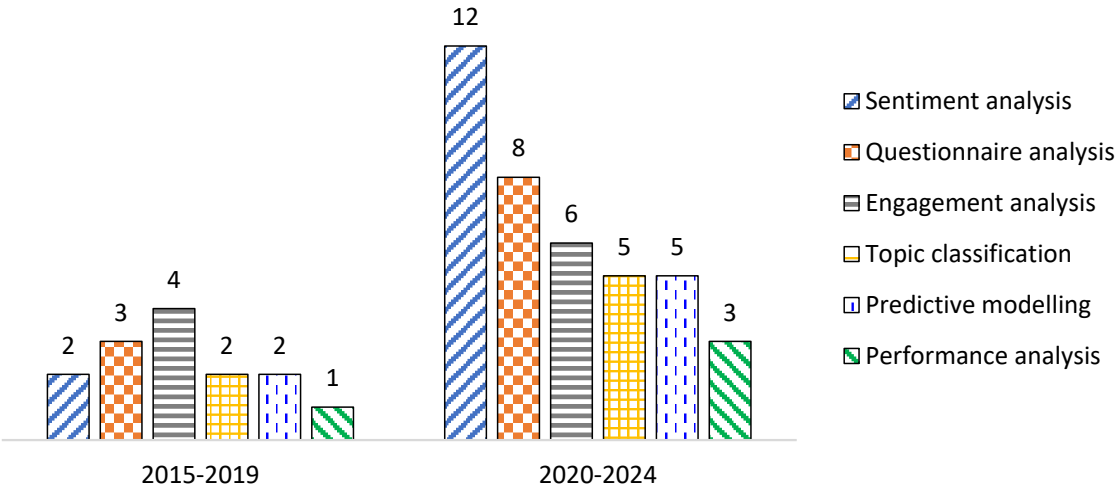


Figure 4. Trends in learning analytics applications across two time periods.

4.1. Sentiment Analysis

Sentiment analysis, also known as opinion mining, examines natural language fragments to identify and classify the emotional attitudes expressed within a text [43,44]. Through a systematic examination of students’ textual feedback, educators can gain insights into their emotional responses and pinpoint possible areas of frustration or confusion. These insights help to create a responsive, engaging, and effective learning environment.

Table 1 summarises the applications of learning analytics reported in the selected studies for sentiment analysis. These studies leverage a variety of data, including MOOC reviews, student reflections, and open-ended feedback from questionnaires. The sample sizes across these studies vary considerably, ranging from as few as 118 open-ended responses from 42 students to over 217,000 course review comments, reflecting the scalability of sentiment analysis techniques across different contexts. Notably, only two studies explicitly specified the evaluated aspects, focusing on educational experiences such as learning content and resources, teaching methods, lecture quality, and assessment. These aspects are fundamental for identifying actionable insights to enhance educational processes.

Table 1. Summary of learning analytics applications for sentiment analysis.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[45]	Students' textual comments	Course reviews on China University MOOC platform	11,293 reviews	N/A	Convolutional Neural Network (CNN), Long Short-Term Memory	Capturing nuanced sentiments can be challenging due to diverse expressions.
[46]	Students' textual comments	Course reviews on China University MOOC platform	217,778 review comments	Course content, teaching methods, lecture, assessment	SBSA leveraging the BERT model, and ABSA employing the LCF-ATEPC model	Current model only identifies negative and positive emotions, missing neutral ones.
[47]	Students' textual comments	Course reviews on China University MOOC platform	9944 reviews	N/A	BERT-BiLSTM-Attention	Using text alone for sentiment analysis limits the ability to capture emotional nuances presented in audio and visual modalities.
[48]	Students' textual comments	MOOC platform (http://open.163.com/)	1850 reviews	N/A	Multi-swarm particle swarm optimisation (MPSO) method	Lack of a specialised sentiment lexicon and appraisal expressions for learners in MOOCs.
[49]	Students' textual comments	Students' reflection questions	42 students, 118 open-ended response sets	N/A	Unsupervised machine learning	• Terms may be inaccurately labelled due to insufficient contextual assessment and reliance on lexicons; • Unsupervised learning lacks depth and accuracy; • The analysis focused on terms with inconsistent sentiment values, limiting the overall scope.
[50]	Students' open-ended feedback	Student questionnaire	1603 phrases	Instructor, lecture, textbook, exams, homework, course management portal, office hours, labs	Google Cloud Platform's Sentiment Analysis API	• Limited user autonomy in terms of input and customisation options; • Transitioning to a web-based format may introduce technical difficulties.

A wide range of analysis methods and tools have been employed in sentiment analysis, with deep learning techniques being the most commonly utilised, especially for the analysis of MOOC reviews. For instance, Refs. [45,46] adopted Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) models, while Ref. [46] also incorporated

BERT and the LCF-ATEPC model for sentiment- and aspect-based sentiment analysis, respectively. Huang and Raga [47] have implemented a BERT-BiLSTM-Attention model, which combines the contextual strengths of BERT with the sequential processing capabilities of BiLSTM and attention mechanisms. **Their model can account for the major components of a text, resulting in a highly robust framework for sentiment analysis.**

Machine learning techniques have been also widely utilised in sentiment analysis. For example, Liu et al. [48] have employed the multi-swarm particle swarm optimisation (MSPSO) method to enhance sentiment recognition in online course reviews. This method reduces feature redundancy and selects discriminative emotional features through optimised feature selection using a binary vector encoding strategy. Roy and Rambo-Hernandez [49] have used unsupervised machine learning methods to uncover patterns and sentiments in student reflections on course activities. Their approach incorporated external references, such as lexicons, to support the sentiment analysis. Additionally, cloud-based tools, such as the Google Cloud Platform’s Sentiment Analysis API, have been applied to analyse sentiments in student feedback, showcasing the integration of scalable, cloud-based services in educational analytics [50].

4.2. Predictive Modelling

Predictive modelling has become a key focus in learning analytics, reflecting the increasing significance of data-driven decision making in educational settings [51]. This approach leverages statistical techniques and machine learning algorithms to analyse data and predict educational outcomes, empowering educators to refine teaching strategies based on actionable insights. Table 2 provides an overview of the various learning analytics applications employed for predictive modelling.

Table 2. Summary of learning analytics applications for predictive modelling.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[52]	Prior knowledge, demographic information, participation and learning variables, academic achievement	University records, LMS, questionnaires	Five courses, 722 students	Overall effectiveness on learning outcomes	Multiple regression analysis, factor analysis	Insufficient assessment of the reliability of affective and motivational items in web-based courses.
[53]	Student demographics, background, course enrolment numbers, course assessment data, experience with online courses, academic records, participation levels in LMS	Student information database, institutional databases, LMS	One course, 214 students	Instructional modalities, student engagement, assessment methods	Machine learning techniques such as random forests and lasso regression	The presence of missing data may affect the results.

Table 2. Cont.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[14]	Student log data, satisfaction scores	MOOC platform, student questionnaire	16 courses, 6016 students, 308,517,712 learning behaviour data	Workload, need fulfilment, intelligibility, instructor style, and student engagement	Deep neural network (DNN) with regression	Lack of consideration for variations in course content and difficulty, which may impact student engagement.
[17]	Students' textual comments	MOOC reviews dataset	66,000 reviews	Learning content and resources	Supervised learning algorithms: naïve Bayes, support vector machines, logistic regression, k-nearest neighbours, and random forest algorithm; ensemble learning methods: AdaBoost, Bagging, Random Subspace, voting, and Stacking	The study emphasises classification accuracy as a primary evaluation metric. Other metrics, such as precision, recall, and F1-score, could provide a more comprehensive assessment of model performance, especially in imbalanced datasets.
[54]	Students' textual comments	Twitter dataset	162,969 tweets	N/A	Bi-LSTM, logistic regression, support vector machine, and naïve Bayes classifier	• Lack of contextual analysis, which can alter sentiment interpretation; • The effectiveness of hyperparameter tuning was not thoroughly explored, possibly limiting model performance insights.
[55]	Students' textual comments	Bangladeshi learning websites	12,536 data points	Quality of instruction, overall course experience	Transformers: XLM-RoBERTa; deep learning methods: CNN and Bi-LSTM, machine learning: naïve Bayes, support vector machine, and random forest, ensemble learning, and hybrid approaches	• Limited dataset diversity restricts comprehensive sentiment analysis across varied contexts; • Challenges arise in accurately analysing sentiments across different languages and cultural contexts.

Table 2. Cont.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[56]	Students' open-ended feedback	Student questionnaire	One course, 280 students, 232 text data points	N/A	Random forest, J48 decision tree, naïve Bayes, and support vector machine	• The highest accuracy achieved was 63.79%, indicating potential limitations in sentiment classification; • Limited dataset.

A range of data types and sources have been utilised in predictive modelling studies. Three studies [14,52,53] relied on quantitative data, including student log data, course assessment scores, and demographic information. These datasets were obtained from platforms such as student information systems, learning management systems, and student questionnaires, and were used to build models that identify key predictors of course effectiveness. In contrast, four studies [17,54–56] focused on qualitative data (e.g., textual comments and open-ended feedback) gathered from MOOC platforms, learning websites, and Twitter datasets, in order to analyse students' sentiments. This diversity in data types and sources enables a comprehensive analysis of both behavioural patterns and subjective experiences.

The evaluation aspects across these studies encompass specific educational outcomes, such as workload and student engagement, as well as broader measures such as instructional quality and overall course experience.

The analysis methods and tools employed for predictive modelling are also diverse. Derr [52] has used statistical methods, including multiple regression analysis and factor analysis, to identify key predictors of academic success and evaluate their impact on learning outcomes. Traditional machine learning techniques, such as random forests and lasso regression, have been employed to predict student performance [53], while deep neural networks (DNNs) with regression—a deep learning technique—have been used to forecast students' course evaluation scores [14].

Machine learning techniques, including naïve Bayes, support vector machines, and random forests, have been widely applied to classify student sentiment into categories such as positive/negative [17,54,56] or positive/negative/neutral [55]. Ensemble methods, which combine multiple models to improve predictive accuracy, have also been employed in relevant studies [17,55]. Advanced deep learning techniques have also been used to analyse sentiment, providing a foundation for the implementation of predictive modelling [54,55].

4.3. Questionnaire Analysis

Questionnaire analysis involves the systematic evaluation of data collected through questionnaires distributed to students, educators, or stakeholders in an educational context. Table 3 summarises the studies that have employed learning analytics—specifically, descriptive analytics methods—to analyse student questionnaire data.

Table 3. Summary of selected learning analytics applications for questionnaire analysis.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[57]	Course rating, teacher rating	Student questionnaire	N/A	Course organisation, content, assessment, workload, learning outcomes, teacher	Descriptive analytics	Its focus on liberal arts limits generalisability to other disciplines.
[58]	Course rating, teacher rating	Student questionnaire	669 courses and 2664 responses	Course organisation, content, assessment, workload, learning outcomes, teacher	Descriptive analytics	- Limited focus on performing arts education restricts broader applicability; - Exclusion of small class data may omit significant insights.
[59]	Quantitative and qualitative feedback	Student questionnaire	228 responses	Overall learning experience, objectives, difficulty of the assignments, material, difficulty level of the course, live session, virtual approach of teaching, performance in the virtual mode of learning	Descriptive analytics	Limited sample size.

Li et al. [57] have focused on **quantitative data related to course and teacher ratings**. Their study evaluated key aspects such as course organisation, content, assessment methods, workload, learning outcomes, and teacher effectiveness. The analysis aimed to identify courses requiring a closer examination to pinpoint areas in need of improvement. Similarly, Ref. [58] analysed a large dataset comprising 669 courses and 2664 student responses. The study focused on comparable evaluation criteria and sought to identify underperforming courses that might benefit from targeted enhancements. Kurni et al. [59] have adopted a mixed-methods approach, incorporating both quantitative and qualitative feedback. Using a five-point Likert scale and open-ended questions, they gathered 228 responses that provided detailed insights into the virtual learning experience. They aimed to uncover factors that could inform the development of effective pedagogical strategies to enhance the quality of online courses.

4.4. Engagement Analysis

Student engagement is a crucial factor that profoundly influences their comprehension, learning experience, and overall performance by the end of a course [60,61]. Therefore, analysing engagement is vital for maintaining quality in the context of higher education [62]. Table 4 illustrates the applications of learning analytics for engagement analysis reported in selected studies. The learning analytics for engagement analysis in course evaluation encompassed three distinct studies, each utilising data from online learning platforms. These studies employed various data types and analysis methods to evaluate different aspects of course effectiveness.

Table 4. Summary of selected learning analytics applications for engagement analysis.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[37]	Previous year's average grades, log data	e-learning platform Open eClass	40 courses, 985 students	Content quality and quantity, overall course effectiveness	Ranking algorithms: CCA (Course Classification Algorithm) and SUGAL (Suggestion Algorithm)	• Validation limited to one Greek university's department affects generalisability of findings; • Some courses, labelled "Overlapped", cannot be ranked in multiple algorithm runs; • S-Algo sometimes fails to cluster or rank low-order courses effectively.
[63]	Course structure, learner demographics, performance data, daily event logs	edX platform	1608 students	Course structure, student engagement, instructional methods, feedback mechanisms, assessment methods	Social network analysis, descriptive analytics	• Findings may not apply to other populations beyond Boeing engineers in workforce training; • Short course duration may limit depth of learning and retention for practical application.
[64]	Quantitative data on behavioural engagement, qualitative data on emotional and cognitive engagement	Moodle, personal meta-reflective diaries (MRDs)	56 students, 9 students' complete MRDs with 27,362 words	Teaching mode, teaching methods	Descriptive analysis, phenomenological analysis with predefined codes	• The number of students included in the data analysis was limited, due to the extensive effort required for in-depth manual analysis; • The study focused solely on log data from Moodle.

Kazanidis et al. [37] have analysed the previous year's average grades and log data from 40 courses involving 985 students. They focused on evaluating content quality and overall course effectiveness using ranking algorithms such as the Course Classification Algorithm and the Suggestion Algorithm. Ginda et al. [63] have incorporated course structure, learner demographics, performance data, and daily event logs from a sample of 1608 students. Social network analysis and descriptive analytics were utilised to assess various aspects of the course, including course structure, student engagement, instructional methods, feedback mechanisms, and assessment methods. Yang and Ghislandi [64] have investigated behavioural, emotional, and cognitive engagement using data collected from Moodle and students' personal meta-reflective diaries. Combining descriptive and phenomenological analyses with pre-defined codes, they conducted an in-depth examination of teaching modes and methods.

4.5. Topic Classification

Topic classification is a widely used method in learning analytics for conducting comprehensive content analysis. This approach involves categorising and organising textual data to identify themes and patterns within educational materials or student feed-

back [65]. Table 5 provides an overview of various learning analytics applications focused on topic classification.

Table 5. Summary of selected learning analytics applications for topic classification.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[15]	Students' textual comments	MOOC course reviews dataset	50 courses	N/A	Latent Dirichlet Allocation (LDA), classification with the naïve Bayes algorithm	There is a need to add model performance metrics, such as inter-word coherence in topic and inter-topic coherence.
[66]	Students' open-ended feedback	Course evaluation survey	36 courses, 742 responses	Workload, teaching methods, learning support	Structural topic model	• Users must download, format, and analyse data externally before re-uploading them; • The study intentionally excluded collaborative tagging to concentrate on specific research issues.
[67]	Students' open-ended feedback	Student questionnaire	16 students	Course material, platform, quizzes and exercises, personal tutor, learning outcomes	Kano model and the machine learning tool Word2Vec	Limited sample size.

Lubis et al. [15] have analysed students' textual comments from a MOOC course reviews dataset, examining 50 courses using the Latent Dirichlet Allocation and naïve Bayes classification algorithms. They extracted different topics and identified the word–topic probability under each topic. Grönberg et al. [66] have focused on students' open-ended feedback collected from a course evaluation survey involving 36 courses and 742 responses. They assessed workload, teaching methods, and learning support through a structural topic model. The proposed tool, called Palaute, effectively identified the most common themes in student feedback and generated indicators of high teaching quality. Marutschke and Hayashi [67] have examined open-ended feedback from a questionnaire involving 16 students, evaluating aspects such as course material and learning outcomes. They employed the Kano model and the machine learning tool Word2Vec to gain insights into student expectations and their overall learning experience.

4.6. Integrated Applications

A total of 12 studies utilised integrated applications of learning analytics for course evaluation, offering a comprehensive framework for assessing the effectiveness of courses and enhancing the overall learning experience. **The most common approaches include the integration of topic classification with sentiment analysis, as well as the combination of engagement analysis with questionnaire analysis.**

As shown in Table 6, the combination of topic classification and sentiment analysis focuses on analysing qualitative data, such as textual comments from course reviews [16,68] and feedback collected through questionnaires [69,70]. This integrated approach aims to

implement content analytics to explore students' perceptions, emotions, and experiences regarding the course content, instructional methods, and interactions.

Table 6. Summary of selected integrated learning analytics applications for topic classification and sentiment analysis.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[16]	Students' textual comments	Course reviews at website https://www.coursereport.com	10,610 reviews	N/A	BERT (Bidirectional Encoder Representations from Transformers), RoBERTa (Robustly optimised BERT approach), and XLNet (Generalised Auto-regression Pre-training)	- The categorisation of topics was not well-focused; - There is potential overlap in terminology between different courses, which could lead to confusion and misinterpretation in the evaluation process.
[68]	Students' textual comments	Chinese MOOC platform called Guokr Web	50 courses, 12,524 reviews	N/A	A machine learning method called Behaviour–Sentiment Topic Mixture (BSTM)	- The need for tailored phrase segmentation and sentiment annotation rules limits applicability across different studies; - Human efforts for textual content labelling may introduce biases and inconsistencies.
[69]	Students' qualitative feedback	Student questionnaire	1067 students	Course content, interactivity, assessment, overall effectiveness	AFINN-111 lexicons, LDA	- Lack of demographic data prevents assessment of the sample's representativeness. - Non-native speakers may have been discouraged from providing feedback or offered less detailed responses.
[70]	Students' qualitative feedback	Student questionnaire	2500 survey responses (625 for each of four open-ended questions)	Course logistics and fit, content and structure, teaching modality, teaching, assessment, resources, peer and teacher interaction	LLMs GPT-4 and GPT-3.5	- Data were only from online biomedical science courses, limiting generalisability; - The study exclusively used English, excluding other languages from analysis; - Results heavily relied on prompting techniques, leading to inconsistent model performance; - Focused primarily on OpenAI models without comparing other advanced models.

With the assistance of deep learning, machine learning methods, and large language models, substantial sample sizes can be effectively analysed. For example, Koufakou [16] has examined 10,610 reviews using advanced models such as BERT, RoBERTa (Robustly optimised BERT approach), and XLNet (Generalised Auto-regression Pre-training) to gain insights into student sentiments and topics. Liu et al. [68] have analysed 50 courses with a total of 12,524 reviews, employing a machine learning technique known as Behaviour–Sentiment Topic Mixture (BSTM) to identify the most frequently discussed topics among learners, analyse the sentiments associated with these topics, and understand how students engage with them. Additionally, in Borakati's study [69], data gathered from 1067 students were analysed through the use of text mining techniques that focused on identifying the most frequently used words. The analysis also included sentiment evaluation using the

AFINN-111 and syuzhet lexicons, along with topic modelling employing Latent Dirichlet Allocation (LDA). Parker et al. [70] have analysed 2500 qualitative survey responses through classification, extraction, thematic analysis, and sentiment analysis. These tasks were executed using the large language models GPT-4 and GPT-3.5, showcasing the models' capabilities in terms of processing and interpreting complex textual data efficiently.

Table 7 presents a summary of learning analytics applications focused on engagement and questionnaire analyses. These applications typically involve data such as students' learning activities, quantitative perceptions of course materials, and interactions with course content, obtained from data sources including platforms such as Moodle and MOOCs, alongside student questionnaires and focus group discussions. The sample sizes vary, with studies often involving hundreds of students across multiple courses (e.g., 659 students in four courses or 3691 log file records from a single course). Evaluation aspects focus on critical elements such as course design, communication effectiveness, workload management, and overall learning experiences. Descriptive statistics were primarily employed in these applications, while data mining techniques were used in Ref. [71] to trace students' activities.

Table 7. Summary of selected integrated learning analytics applications for engagement and questionnaire analyses.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[71]	Students' learning activities and quantitative perceptions of the course	Moodle, student questionnaire	Four courses, 659 students, 260 questionnaires	Course design, contents and assignments, communication, workload, overall learning experience	Data mining, descriptive statistics	- Limited sample size; - The lack of analysis on instructor characteristics leaves a gap in understanding their impact on course outcomes.
[72]	Students' perceptions, "posted messages" in the Moodle platform	Students' questionnaire survey, focus group discussion, Moodle	One course, 18 students	Course organisation, resources, assignments, acquired competences	Descriptive statistics	- Limited sample size; - The survey was conducted by the teacher, which introduces a risk of bias.
[73]	Student background information, interactions with course materials, progress through the course modules, students' feedback, discussion forum contributions	FutureLearn MOOC platform, surveys conducted at various points during the course	380 students	Course content and materials, learning activities	Descriptive analytics	The course's high attrition rates limit the generalisability of findings, as only a small subset of engaged learners provided feedback.
[74]	Course information, students' behaviour, quantitative perceptions of the course, level of satisfaction with the course	Moodle, student questionnaire	One course, 3691 log file records	Organisation and presentation, learning objectives and assessments, interpersonal interaction, use of technology, course materials	Descriptive analytics	Limited number of courses at only one higher education institution.

As shown in Table 8, one study [12] integrated performance and questionnaire analyses for course evaluation. The authors embedded a learning analytics tool within the learning

management system to enhance advising and support throughout the academic year. Unlike the traditional reliance on grades, this tool provides real-time data to multiple stakeholders, facilitating meaningful discussions about learning objectives and identifying areas for course improvement.

Table 8. Summary of a study utilising integrated learning analytics application for performance and questionnaire analyses.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[12]	Course objectives with corresponding overall student achievement percentages, students' quantitative perceptions of the course and instructors	Desire2Learn Brightspace LMS, student questionnaire	50 students	Teaching styles, course content, alignment between learning objectives and assessments	Descriptive analysis	- Limited sample size; - Lack of qualitative insights.

Table 9 summarises the integrated learning analytics applications used for engagement, questionnaire, and performance analyses in the context of course evaluation. Notably, all three studies were conducted during the COVID-19 pandemic, focusing on the transition from face-to-face to online course delivery. These applications align with other research indicating that student engagement, performance, and perceptions are crucial elements of online course experiences [75,76]. While learning analytics were applied for various purposes, only descriptive analysis was utilised across these studies.

Table 9. Summary of selected integrated learning analytics applications for engagement, questionnaire, and performance analyses.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[13]	Student performance, interactions with course materials and activities on LMS, evaluations of the course	Academic records, LMS, student questionnaire	Six courses from each of two academic years, 300 students	Content quality, teaching methods, skills development, student involvement, curriculum structure, feedback mechanisms	Descriptive analysis	- Limited sample size; - Findings from the emergency remote teaching context may not be generalisable to well-planned online learning environments.
[77]	Student performance, satisfaction and engagement/behaviour assessment results, student evaluations of learning and teaching, learning activities	Canvas, student questionnaire	98 learners	Learning objectives, course organisation, strategies, resources and technologies, assessment methods, feedback, overall satisfaction	Descriptive analytics	- The study's findings are limited to the Foundations of Wine Science course, reducing their generalisability to other subjects or disciplines. - The research examines the shift from face-to-face to online learning, which may introduce confounding factors affecting student experiences.

Table 9. Cont.

Ref.	Data Type	Data Source	Sample Size	Evaluation Aspect	Analysis Method/Tool	Limitation
[78]	Students' performance, behaviour, student opinions from surveys, interaction data	Course management system, Google Analytics and Weebly Analytics, YouTube Analytics, student questionnaire	52 students	Learning outcomes, technology integration, course structure	Descriptive analytics	Lack of consideration for technology access, support resources, and family context.

5. Discussion

The findings of this study illuminate the features and trends associated with the application of learning analytics for course evaluation. They contribute to advancing practices using learning analytics to assess course effectiveness, providing insights into the current state of development and identifying future research needs in this area. The research questions of this study are revisited and addressed below, in accordance with the insights derived from the findings.

Research question 1: What are the application categories of learning analytics in course evaluation?

In the realm of learning analytics for course evaluation, six primary application categories were identified, with sentiment analysis emerging as the most prominent. Its usage has seen a significant increase from 2020 to 2024, underscoring its growing importance in capturing and interpreting student emotions and opinions to enhance course effectiveness. This surge can be attributed to advancements in cutting-edge AI algorithms during this period [79]. However, sentiment analysis is not without challenges, including difficulties in interpreting context, sarcasm, complex language, and mixed sentiments, which can lead to misclassification and inaccuracies in feedback analysis [44,80].

Following sentiment analysis is questionnaire analysis, with three and eight instances recorded in the two periods, respectively. This demonstrates its continued relevance as a tool for gathering structured feedback from students. Engagement analysis ranked next (with 10 instances in total), highlighting the critical role of assessing student interaction and participation in course evaluation. Notably, engagement analysis was the leading application during the 2015–2019 period, likely due to the limited technological advancements at that time. Metrics such as attendance, participation, and activity tracking are more straightforward to measure and analyse, compared to the complexities of sentiment analysis, making engagement analysis foundational in the early stages of learning analytics.

Topic classification and predictive modelling each accounted for seven instances, showcasing their utility in organising qualitative feedback and forecasting course effectiveness, respectively. Finally, performance analysis was the least prevalent category, typically used in conjunction with other applications. This integration offers a more comprehensive understanding of student outcomes and complements insights gained from engagement and questionnaire analysis.

Research question 2: What are the features of learning analytics in these practices, in terms of data collection methods, analysis techniques, and evaluated aspects?

The most common data source for course evaluation is student questionnaires, which collect both qualitative data through open-ended questions and quantitative data via Likert scales or ratings for courses and instructors. Typically administered at the end of a course, student questionnaires provide valuable insights; however, it is worth noting that only one study conducted surveys at various points during the course [73]. In addition to questionnaires, learning management systems, MOOC platforms, and educational

websites serve as common data sources. These platforms provide information on learner demographics, behaviours, and performance. The accessibility of these data has increased with the rise of online learning, offering more objective perspectives for course evaluation.

In terms of learning analytics techniques, descriptive analysis was the most commonly used, particularly for questionnaire, engagement, and performance analyses. More advanced techniques, such as machine learning approaches, have been employed for tasks including sentiment analysis, topic classification, and predictive modelling. Emerging artificial intelligence tools, including large language models such as GPT-4 and GPT-3.5, have been utilised for classification, extraction, thematic analysis, and sentiment analysis of qualitative feedback. Additionally, social network analysis has been carried out to examine student engagement.

Course evaluation encompasses several critical aspects that contribute to the overall learning experience and effectiveness, aligning with the primary goals of evaluation. Key elements include course design, content quality, assignments, and instructional methods, all of which significantly influence student engagement and learning outcomes. Factors such as effective communication, manageable workload, and the organisation of course materials are evaluated to foster a positive educational environment. Moreover, assessment methods and feedback mechanisms are monitored to track student progress and ensure alignment with learning objectives, while the integration of technology and interactive resources is assessed to enhance the online learning experience. Typically, the aspects evaluated are specified through structured student questionnaires or the intentional collection of behavioural data via learning platforms. In contrast, students' textual comments are generally provided freely, without specific guidance, which means they may not clearly specify the evaluated aspects. While this unstructured feedback allows for rich, nuanced insights, it can also introduce ambiguity and make it challenging to derive actionable conclusions [81].

Research question 3: What are the main limitations in existing studies focused on learning analytics in course evaluation?

The studies examining the application of learning analytics in course evaluation revealed several key limitations that hinder the robustness and generalisability of their findings. One prominent issue is limited sample sizes, which often restrict the ability to draw broad conclusions that are applicable to diverse populations. Additionally, the reliance on surveys conducted by instructors introduces potential bias, as students might tailor their feedback based on perceptions of the instructor's expectations. Moreover, while lexicon-based approaches are critical for sentiment analysis, the absence of a specialised sentiment lexicon for student review comments hampers the effectiveness of this method. This gap can lead to insufficient contextual analysis and inaccuracies in sentiment interpretation, resulting in inconsistencies across different studies. Furthermore, the focus on specific subjects—such as liberal arts or biomedical sciences—limits the applicability of the obtained findings to other disciplines, potentially overlooking valuable insights from a broader educational context.

Another significant limitation is the reliance on quantitative measures, which may not fully capture the complexities of student experiences. While descriptive and advanced analytical techniques such as machine learning offer valuable insights, they often fall short in addressing the nuanced emotional responses of learners. Sentiment analysis frequently categorises text into just two or three sentiment types, reducing the dimensionality of the data and potentially overlooking more meaningful details. Additionally, challenges in analysing sentiments across different languages and cultural contexts further complicate the evaluation process. Collectively, these limitations underscore the need for more com-

prehensive and inclusive approaches in future research focused on learning analytics for course evaluation.

6. Conclusions

This study contributes to the growing body of literature on learning analytics by systematically examining its applications in course evaluation. By adhering to the PRISMA guidelines, the research ensured a rigorous and transparent methodology, allowing for a comprehensive synthesis of relevant studies. The identification of six primary application categories—including sentiment analysis, questionnaire analysis, engagement analysis, topic classification, predictive modelling, and performance analysis—highlights the diversity of approaches employed in the evaluation of course effectiveness. Furthermore, the exploration of data collection methods and analytics techniques provided insights into how learning analytics can be leveraged to enhance educational practices. Overall, the findings of this research underscore the significance of learning analytics in understanding student experiences and improving course design, ultimately fostering better learning outcomes.

The findings of this study offer implications for educators and institutions seeking to implement learning analytics effectively. By revealing the predominant reliance on student questionnaires and the increasing prominence of sentiment analysis, this study emphasises the need for institutions to adopt more nuanced approaches to data collection and analysis. Additionally, the insights gained through the evaluation of different learning analytics techniques can guide educators in selecting appropriate methods for their specific contexts. The study also highlighted the importance of addressing the limitations associated with current research, such as the need for specialised sentiment lexicons and the exploration of qualitative data, in order to enhance the robustness and applicability of the obtained findings to diverse educational settings.

The present study was limited by its reliance on a specific timeframe and database, focusing exclusively on articles published within the past decade in Scopus. This may restrict the breadth of perspectives considered, potentially overlooking insights from other databases or earlier studies. Additionally, the stringent inclusion criteria may have led to the exclusion of some relevant studies, thus affecting the comprehensiveness of the review. Addressing these limitations in future research could provide a more holistic understanding of the application of learning analytics across different educational settings and populations.

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